

LOADZONE · DESIGN DECISION RECORD

LoadZone recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	LDZ-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/loadzone		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	01 decided by Amish on 2026-10-02 as recommended in LDZ-DEC-001
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below marked "Decided" is decided by Amish, 2026-09-25: go with recommendation. The item without a recommendation (O1) was given one in the design decisions register (LDZ-DEC-001, item 2) and decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions."

■ **Context**

After the TRL 3 session, LoadZone had eight items adopted for TRL 3 work pending Amish's review (LDZ-DDR-001, D1 to D8) and six new items raised at TRL 3 (O2 to O7), each with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Where a recommendation offered several options, the recommended option is the decision. Items that carried no recommendation are not decided by this record.

■ **Decision**

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1 to D8	LDZ-DDR-001 items: surface-bonded puck; magnetometer only; one puck per 7 m slot; two AA-size Li-SOCl ₂ cells; cast polyurethane dome over fully potted electronics; sign outside the \$120 budget, which covers the two-puck kit only; LoRaWAN class C downlink to the sign; a feed that maps onto the Curb Data Specification	As recommended in LDZ-DDR-001	LDZ-DDR-001 v0.2 status wording; LDZ-PRC-001 v0.4 "Key design choices" now decided. <code>budget_usd</code> stays \$120 (D6 redefined the budget's scope rather than its figure); pitch and problem unchanged, since no rewording was recommended
O2	R4 link from under a van	Option (a): restate R4 as a gateway within 300 m of the bay	R4 in LDZ-REQ-001 v0.4 now reads 300 m, not 1 km; LDZ-CAL-001 v0.2 [D3]: 353 m range at SF9 with a 10 dB fade margin, 1.18 times the distance. R4 moves from not met to met on paper. Precis, problem statement, BOM notes, blueprint key figures and a note on LDZ-DWG-001 Rev P2
O3	R6 bond under braking	Option (c): two-part road-marker epoxy, and R6 restated with the finding that parked wheels straddle the puck	BOM line 7 is now a two-part road-marker epoxy bed (was a bitumen pad or epoxy), price unchanged at \$4.00 per puck; <code>cad/src/model.py</code> comment, drawing material note and a bond note on LDZ-DWG-001 Rev P2; exploded-view label; R6 restated around a maneuvering wheel crossing the puck. R6 stays at risk: bond shear 1.51 MPa against about 1.0 MPa assumed, factor 0.66 (unchanged). The pull-off test is TRL 4 work and on hold
O4	R11 night legibility	Option (a): a daylight-only legibility target	R11 restated to 25 m by day; about 28 m by day on paper, so R11 moves from not met (night) to met on paper
O5	Sign network	The sign option requires a private network server (TwinKit) or a city network that allows about 200 downlinks a day; on The Things Network it would need LoRa point to point	Stated as a decided constraint in the precis and problem statement; TwinKit class C support listed as a cross-repo action
O6	Engineering proposals	Confirmed: one Schottky diode per cell, an 85 °C hybrid pulse capacitor, a fully potted cavity and a 6 mm crown radius	Already in the model and BOM; wording changed from "awaiting confirmation" to decided. No geometry or cost change
O7	Payload size	Pack the payload into 11 bytes if the band chosen in O1 is US915	Firmware rule stated in the precis and requirements; LDZ-CAL-001 v0.2 [A2]: 370.7 ms at SF10 for 11 bytes. Figures stay on EU868 until the band is chosen. Writing the firmware is TRL 4 work and on hold

LDZ-CAL-001 v0.2 also takes FieldNode's published 100 mW sensor allowance (FND-DDR-002) in place of the earlier 115 mW and 100 mW pair; the sign's 17.7 mW is 18 % of it, and R12 stays met on paper.

Table 2. Item left open by this record, decided on 2026-10-02.

#	Item	Status
O1	First trial partner (a city curb team, a business district or a carrier) and the LoRaWAN band. No recommendation was made.	Decided by Amish, 2026-10-02: first candidate type to approach (not yet agreed), a city curb-management or transportation team that has published or piloted the Curb Data Specification, in a city without routine snow plowing; US915 is the default band, since the 11-byte payload already fits it (LDZ-DEC-001, item 2)

The suggestion in `docs/REVIEW.md` that CurbCount and LoadZone could share one server and data feed through CityTwin was a suggestion, not a recommendation awaiting decision, and is not adopted.

■ Consequences

- Requirement status (LDZ-CAL-001 v0.2): none not met, 3 at risk (R1, R5, R6), 1 not verifiable at TRL 3 (R14), 7 met on paper, 4 met by design. Before: 2 not met (R4, R11), 3 at risk, 1 not verifiable, 5 met on paper, 4 met by design.
- Controlled documents revised: LDZ-PRB-001 v0.4, LDZ-PRC-001 v0.4, LDZ-REQ-001 v0.4, LDZ-CAL-001 v0.2, LDZ-DDR-001 v0.2; drawing LDZ-DWG-001 Rev P2 (notes and material; geometry unchanged).
- Cost: two-puck kit \$114.00 against the unchanged \$120 `budget_usd` ; sign option \$97.00 plus a host FieldNode at \$126.00, outside the budget.
- Cross-repo actions (other repos not edited): see `docs/REVIEW.md` , session "recommendations accepted".
- TRL 4 remains on hold by Amish's instruction. The bond pull-off and shear test, the field link survey, the magnetometer field log and the firmware payload rule are TRL 4 work and have not been started.